## 1 Label-free graph representation

**Input graph**
$G = (V, E, X)$

**Node features X**

**Adjacency A**

| | $v_1$ | $v_2$ | ⋯ | $v_n$ |
|---|---|---|---|---|
| $v_1$ | 0 | 1 | ⋯ | 0 |
| $v_2$ | 1 | 0 | ⋯ | 1 |
| ⋯ | ⋮ | ⋮ | ⋱ | ⋮ |
| $v_n$ | 0 | 1 | ⋯ | 0 |

**Two augmented views**

edge dropping + feature masking

independent augmentation

**Shared GRACE encoder**
$g_\psi$

**Embedding $H^1$**

**Embedding $H^2$**

**Frozen normalized representation $\bar{H}$**

after training: $\bar{H} = \text{RowNorm}[g_\psi(X,A)]$

## 2 Label-agnostic granular-ball partition

**Geometry-only construction in frozen embedding space $\bar{H}$**

**Initialization**
$B_0 = V$

**Recursive bi-partition**

**Further recursive splitting**

**Embedding space $\bar{H}$ covered by granular balls $\mathcal{B} = \{B_j\}_{j=1}^{K}$**

**selected ball $B_j$**

fixed budget: $K \le B_{max} = 64$

Balls partition all N nodes; only labeled training nodes vote. Construction is label-agnostic; circle overlaps are projection artifacts.

## 3 Leave-one-out ball-level label consensus

**(a) Select a granular ball $B_j$**

**(b) Leave-one-out consensus**

**$p_{j,-v}$ (exclude v)**

v's own vote removed

$$q_{j,-v} = (1-\beta)p_{j,-v} + \beta e_{m_{j,-v}}$$

If the majority ties: $q_{j,-v} = p_{j,-v}$
(tie-safe; no arbitrary one-hot class)

**(c) Construct a soft target $t_v$**

noisy one-hot + local consensus = soft target $t_v$

$$t_v = (1 - \alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v} q_{j,-v}$$

$\alpha_{j,-v}$ measures vote concentration
$\beta = 0.5$ in the fixed configuration

## 4 X-only MLP training and inference

**Training with graph-derived targets**
for each training node $v \in V_L$

**2-layer MLP $f_\theta$**

**node feature $x_v$**

precomputed $t_v$

**Soft cross-entropy** prediction vs. $t_v$ (Eq. 13)

**Inference after training (transductive)**

**trained MLP**

**node feature x**

class probabilities
**no graph access**

Class 1 · Class 2 · Class 3 · Class 4 · Class 5 · non-voting node · excluded self-vote · data flow · contrastive alignment

**X-only inference:** no A, no $\bar{H}$, no neighbors